

Introduction to probability, statistics and data handling

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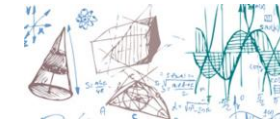
based on lectures by Tomasz Szumlak

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More on Confidence Interval and Hypothesis Testing
(test of two samples)
including
p value and power of a test

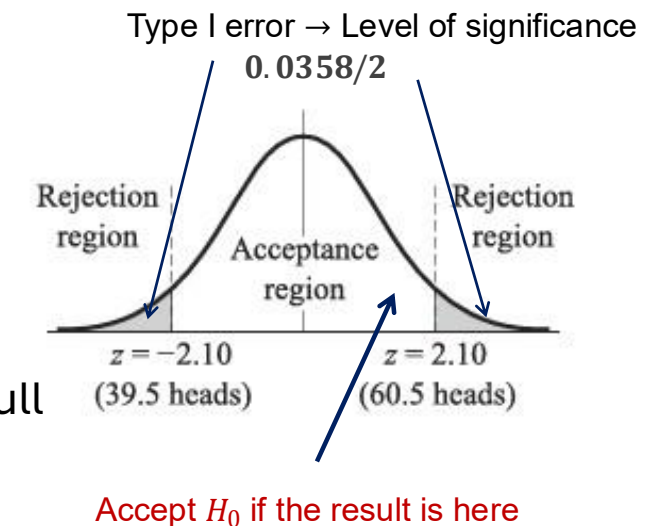
Tests with normal distribution



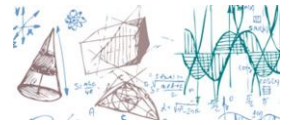
- **So, how to reject the test?** If for a sample, its z-score corresponding to the value of S is outside the 95% interval, we could conclude that this is **unlikely event** that would occur with probability of 5%, if our null hypothesis were true
- In other words, we would say, that the z-score for this sample differed significantly from our expectations based on the H_0 and it should be rejected!
 - ✓ The grey area represents the **level of significance** of the test, i.e., the probability of **Type I error**

Or, we say, that the hypothesis was rejected at 5% level of significance

- All z-scores outside -1.96 and 1.96 are called **critical region** of the rejection of the null hypothesis (the region of significance)
- Not too hard to notice that all z-scores inside this interval will be called the **region of acceptance** of the hypothesis or the region of non-significance.
- So far we can conduct hypothesis test on single means and single proportions.



Type I and Type II errors



- Imagine that we rejected a hypothesis and it happens to be true. This kind of error is called **Type I** and the probability is described by α .
- If we accepted a hypothesis which is not true, we say we made **Type II** error and the probability is described by β .
- In both cases we failed and error in judgement was made, so we'd like to have these values low.

<i>Truth</i>	<i>Decision made</i>	
	Ho not rejected	Ho rejected
Ho is true	Correct (1- α)	Type 1 error (α)
Ho is false	Type II error (β)	Correct (1- β)

α and β should be as small as possible

Type I and Type II errors

Example: A certain experimental drug claims a cure rate of at least 75% for males with prostate cancer. Describe both the Type I and Type II errors in context. Which error is the more serious?

Type I: A cancer patient believes the cure rate for the drug is less than 75% when it actually is at least 75%.

Type II: A cancer patient believes the experimental drug has at least a 75% cure rate when it has a cure rate that is less than 75%.

The Type II error contains the more severe consequence.

If a patient believes the drug works at least 75% of the time, this most likely will influence the patient's (and doctor's) choice about whether to use the drug as a treatment option.

p-value and rare events

It can happen that a rare event can occur – how to calculate the probability then?

p-value:

this is the probability that, **assuming the null hypothesis is true**, the results from another randomly selected sample will be as **extreme or more extreme** as the results from the given sample.

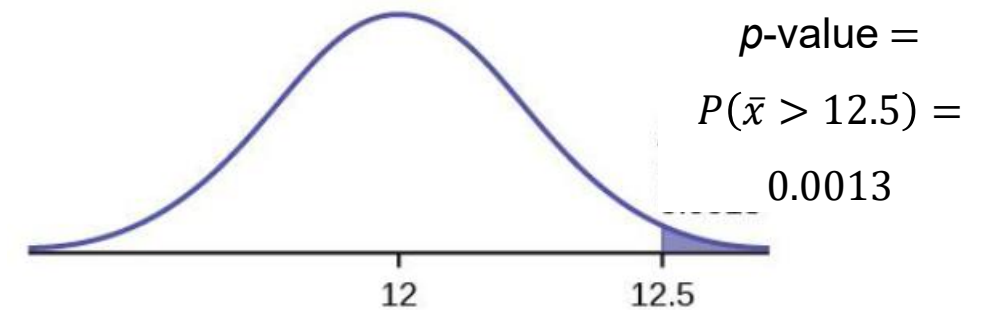
- A large p -value calculated from the data indicates that we should not reject the **null hypothesis**.
- The smaller the p -value, the more unlikely the outcome, and the stronger the evidence is against the null hypothesis.
- We would reject the null hypothesis if the evidence is strongly against it.

Ex: normal distribution has a standard deviation of 1. We want to verify a claim that the mean is greater than 12.

A sample of 36 is taken with a sample mean of 12.5.

$$H_0: \mu \leq 12$$

$$H_a: \mu > 12$$



New decision rule

Remember α ? Type I error? (rejecting the null hypothesis when the null hypothesis is true).

When you make a **decision** to reject or not reject H_0 , do as follows:

- If $\alpha > p\text{-value}$, reject H_0 . The results of the sample data are significant. There is sufficient evidence to conclude that H_0 is an incorrect belief and that the **alternative hypothesis**, H_a , may be correct.
- If $\alpha \leq p\text{-value}$, do not reject H_0 . The results of the sample data are not significant. There is not sufficient evidence to conclude that the alternative hypothesis, H_a , may be correct.
- When you "do not reject H_0 ", it does not mean that you should believe that H_0 is true. It simply means that the sample data have **failed** to provide sufficient evidence to cast serious doubt about the truthfulness of H_0 .

Reject the null hypothesis when **the $p\text{-value}$ is less than the established alpha value**. The results of the sample data **support the alternative hypothesis**.

Do not reject the null hypothesis when **the $p\text{-value}$ is greater than the established alpha value**. The results of the sample data **do not support the alternative hypothesis**.

Two population means

- We want to compare two groups – two means or two proportions
- The two independent samples are simple random samples from two distinct populations.
- For the two distinct populations:
 - ✓ if the sample sizes are small, the distributions are important (should be normal)
 - ✓ if the sample sizes are large, the distributions are not important (need not be normal)

$$\left. \begin{aligned} \bar{X}_1 &\rightarrow N\left(\mu, \frac{\sigma_1}{\sqrt{n_1}}\right) \\ \bar{X}_2 &\rightarrow N\left(\mu, \frac{\sigma_2}{\sqrt{n_2}}\right) \end{aligned} \right\} \begin{aligned} \bar{X}_1 - \bar{X}_2 &\rightarrow N(0, \sigma(\bar{X}_1 - \bar{X}_2)) \\ \sigma(\bar{X}_1 - \bar{X}_2) &= \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \end{aligned}$$

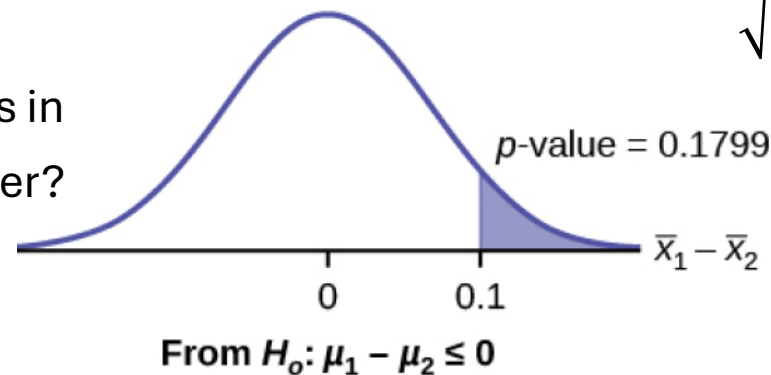
tests statistics:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_{\bar{X}_1} - \mu_{\bar{X}_2})}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Ex: compare two means (incomes in two factories). Is the second higher?

$$H_0: \mu_1 \leq \mu_2$$

$$H_a: \mu_1 > \mu_2$$



Hypothesis – differences of means

- **Ex.** Two classes attended the same course and took an exam. The mean mark for the first class (40 students) was 74 points with a standard deviation of 8, the second (50 students) scored the mean of 78 points with the standard deviation of 7. Can we claim that one of the class is significantly better than other? We set the $\alpha = 0.05$

- ✓ $H_0: \mu_1 = \mu_2$ - no significant difference, just fluctuation
- ✓ $H_a: \mu_1 \neq \mu_2$ - the second class is better
- ✓ The test statistics is the difference of means

$$\mu_{\bar{X}_1 - \bar{X}_2} = 0, \sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{64}{40} + \frac{49}{50}} = 1.606$$

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_{\bar{X}_1 - \bar{X}_2})}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{74 - 78}{1.606} = -2.49$$

we can use z-scores for normal distribution (big sample)

$$z_{1,2} = \pm 1.96 \rightarrow H_0 \text{ rejected @ 5\%}$$

- ✓ p-value: $P(Z \leq -2.49) + P(Z \geq 2.49) = 0.0128$

Examples – C.I. for the difference in means

- **Ex.** A sample of 150 brand A light bulbs showed a mean life-time of 1400 hours and a standard dev. of 120 h. A sample of 200 brand B light bulbs showed the corresponding values of 1200 h and 80 h. Find (a) 95% and (b) 99% *C.I.* for the difference of the mean life-times of the populations of brands A and B.

a) The confidence interval for the difference in means can be written as:

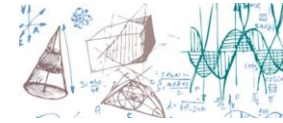
$$\bar{L}_A - \bar{L}_B \pm z_c \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}} = 1400 - 1200 \pm 1.96 \sqrt{\frac{120^2}{150} + \frac{80^2}{100}} = 200 \pm 24.8 \text{ h}$$

We can be 95% confident that the difference in population means lies between 175 and 225 h

b) And for the 99% *C.I.*:

$$\bar{L}_A - \bar{L}_B \pm z_c \sqrt{\frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B}} = 200 \pm 1.96 \sqrt{\frac{120^2}{150} + \frac{80^2}{100}} = 200 \pm 32.6 \text{ h}$$

CI – differences of means



- What would be a change if σ is unknown?
- The test statistics is the same: the difference of means: $\bar{X}_1 - \bar{X}_2 \rightarrow N(0, \sigma(\bar{X}_1 - \bar{X}_2))$
 - but now $\sigma(\bar{X}_1 - \bar{X}_2)$ is estimated:

$$\sigma(\bar{X}_1 - \bar{X}_2) \rightarrow \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$s_1^2 = \sum_{i=1}^{n_1} (X_i - \bar{X})^2 \frac{1}{n_1 - 1}$$

$$s_2^2 = \sum_{i=1}^{n_2} (X_i - \bar{X})^2 \frac{1}{n_2 - 1}$$

- and test statistic is t-score:

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_{\bar{X}_1} - \mu_{\bar{X}_2})}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

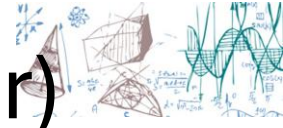
$$\nu = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1} \frac{s_1^2}{n_1} + \frac{1}{n_2 - 1} \frac{s_2^2}{n_2}}$$

- and we need to use t -Student distribution with ν degrees of freedom

- The test comparing two independent population means with unknown and possibly unequal population standard deviations is called the Aspin-Welch t-test.



Examples – C.I. – proportion (reminder)



- **Ex.** A sample poll of 100 voters has been chosen at random in a given district. The result indicated that 55% of them were supporting a party A. Find (a) 95% and (b) 99% c.i. for the proportion of all voters supporting this party.

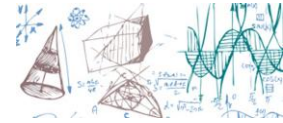
- a) The c.i. for the population p is defined as: $P \pm z_c \sigma_P = P \pm z_c \sqrt{\frac{p(1-p)}{n}}$. In order to estimate parameter p we use the sample proportion (measurement!)

$$P \pm z_{0.975} \sqrt{\frac{p(1-p)}{n}} = 0.55 \pm 1.96 \sqrt{\frac{0.55 \cdot 0.45}{100}} = 0.55 \pm 0.10$$

- b) And the 99% c.i.:

$$P \pm z_{0.995} \sqrt{\frac{p(1-p)}{n}} = 0.55 \pm 2.58 \sqrt{\frac{0.55 \cdot 0.45}{100}} = 0.55 \pm 0.13$$

Examples – C.I. – two proportions



- **How we can compare two independent Population Proportion?**

- There are a few assumptions that must be fulfilled:

- samples should be random and independent

- number of successes > 5

- ✓ This enables using the normal dist, approximation.

- ✓ The pooled proportion RV:

$$P = \frac{x_A + x_B}{n_A + n_B}$$

- ✓ the difference:

$$\Delta_P = P'_A - P'_B$$

$$\Delta_P \rightarrow N(0, \sigma_\Delta)$$

$$\sigma_\Delta = \sqrt{P(1 - P) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}$$

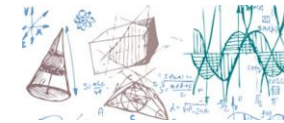
- ✓ and test statistics (z-score):

- $H_0: P_A = P_B$, so $\Delta_P = 0$

- $H_a: P_A \neq P_B$, so $\Delta_P \neq 0$

$$Z = \frac{(P'_A - P'_B) - (P_A - P_B)}{\sqrt{P(1 - P) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}$$

Test of two proportions



Ex. Two types of valves are being tested to determine if there is a difference in pressure tolerances. Fifteen out of a random sample of 100 of Valve *A* cracked under 4,500 psi. Six out of a random sample of 100 of Valve *B* cracked under 4,500 psi. Test at a 5% level of significance.

$$P = \frac{x_A + x_B}{n_A + n_B} = \frac{15 + 6}{100 + 100} = 0.105 \quad P'_A = \frac{15}{100} \quad P'_B = \frac{6}{100}$$

the difference:

$$\Delta_P = P'_A - P'_B = 0.21 \quad \Delta_P \rightarrow N(0, \sigma_\Delta)$$

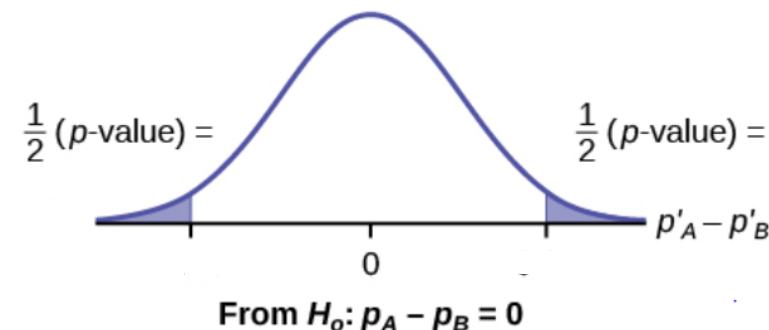
and test statistics (z-score):

$$Z = \frac{(P'_A - P'_B) - (P_A - P_B)}{\sqrt{P(1 - P) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}}$$

$$\sigma_\Delta = \sqrt{P(1 - P) \left(\frac{1}{n_A} + \frac{1}{n_B} \right)}$$

$$H_0: P_A = P_B, \text{ so } \Delta_P = 0$$

$$H_a: P_A \neq P_B, \text{ so } \Delta_P \neq 0$$



The *p*-value is 0.0379, so we can reject the null hypothesis. At the 5% significance level, the data support that there is a difference in the pressure tolerances between the two valves.

Examples – C.I. for variance

$$\chi^2_{0.025} \leq \frac{(n-1)\hat{S}^2}{\sigma^2} \leq \chi^2_{0.975}$$

- **Ex.** The standard deviation of heights of 16 male students chosen at random in a school of 1000 male students yielded: $s = 5.9 \text{ cm}$. Find (a) 95% and (b) 99% c.i. of the standard deviation for all male students at this school. Assume that height is normally distributed.

✓ The critical values in this case are: $S \frac{\sqrt{n}}{\chi_{0.975}}$ and $S \frac{\sqrt{n}}{\chi_{0.025}}$. From a table for $\nu = 15$:

$$\chi^2_{0.975} = 27.5 \rightarrow \chi_{0.975} = 5.24 \text{ and } \chi^2_{0.025} = 6.26 \rightarrow \chi_{0.975} = 2.50$$

$$C.I.^{(s)}_{0.975} = S \frac{\sqrt{n}}{\chi_{0.025}} < s < S \frac{\sqrt{n}}{\chi_{0.975}} = 4.9 < s < 9.4 \text{ cm}$$

✓ The 99% c.i.:

$$C.I.^{(s)}_{0.975} = S \frac{\sqrt{n}}{\chi_{0.005}} < s < S \frac{\sqrt{n}}{\chi_{0.995}} = 4.1 < s < 11.0 \text{ cm}$$

Examples – test of variance

$$\chi^2_{0.025} \leq \frac{(n-1)\hat{S}^2}{\sigma^2} \leq \chi^2_{0.975}$$

Ex. With individual lines at its various windows, a post office finds that the standard deviation for normally distributed waiting times for customers on Friday afternoon is 7.2 minutes. The post office experiments with a single, main waiting line and finds that for a random sample of 25 customers, the waiting times for customers have a standard deviation of 3.5 minutes.

With a significance level of 5%, test the claim that **a single line causes lower variation among waiting Times (shorter waiting times) for customers.**

Ex. A SCUBA instructor wants to record the collective depths each of his students dives during their checkout. He is interested in how the depths vary, even though everyone should have been at the same depth. He believes the standard deviation is three feet. His assistant thinks the standard deviation is less than three feet. If the instructor were to conduct a test, what would the null and alternative hypotheses be?

Examples – test of variance

$$\chi^2_{0.025} \leq \frac{(n-1)\hat{S}^2}{\sigma^2} \leq \chi^2_{0.975}$$

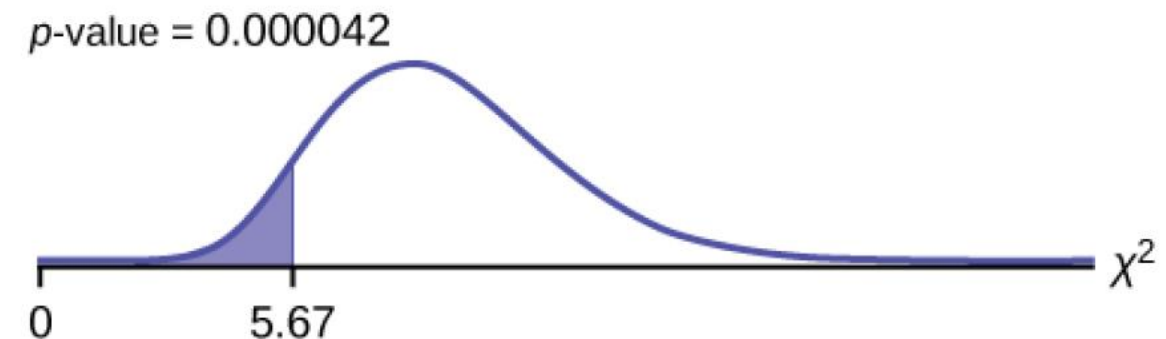
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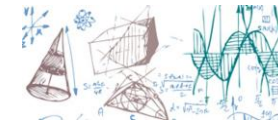
- $H_0: \sigma^2 = 7.2^2$
- $H_a: \sigma^2 < 7.2^2$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(25-1)(3.5)^2}{7.2^2} = 5.67$$

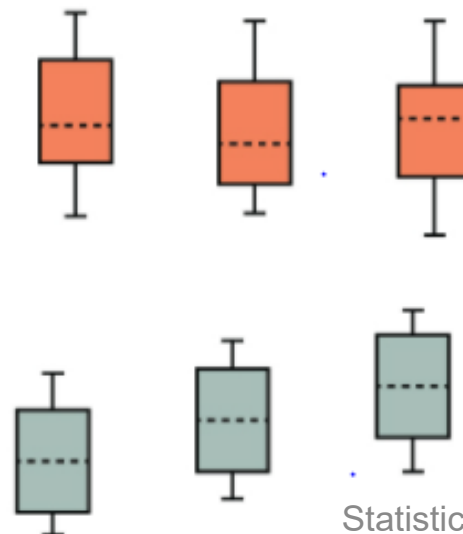
$$\chi^2 = \frac{(n-1)s^2}{\sigma^2} = \frac{(25-1)(3.5)^2}{7.2^2} = 5.67$$



Analysis of variance



- In case of comparing the mean values of two populations we introduced how the sampling distribution of differences (and sums...) of means can be obtained. And what about the distribution of **differences of variances** that can be denote as: $S_1^2 - S_2^2$?
- For hypothesis tests comparing averages between more than two groups: "Analysis of Variance" (ANOVA).
- The null hypothesis is simply that all the group population means are the same. The alternative hypothesis is that at least one pair of means is different.



$H_0: \mu_1 = \mu_2 = \mu_3$ and the three populations have the same distribution if the null hypothesis is true. Differences are due to random variation.

If the null hypothesis is false, then the variance of the combined data is larger which is caused by the different means.

Differences are too large to be due to random variation.

C.I. for variance ratios

- In case of comparing the mean values of two populations we introduced how the sampling distribution of differences (and sums...) of means can be obtained. And what about the distribution of **differences of variances** that can be denote as: $S_1^2 - S_2^2$?
- It turns out, that such distribution is **surprisingly** troublesome! Instead we can use **another statistic**: S_1^2/S_2^2 . We conclude that if the ratio is small/large the difference between variance is large, conversely if the ratio is close to unity the difference should be small, and we have a theorem...
- **Theorem 1.** Consider, we have two random and independent samples, of size n and m respectively, drawn from two normal populations with variances σ_1^2 and σ_2^2 . It can be shown that the statistic

$$F = \frac{\frac{mS_1^2}{(m-1)\sigma_1^2}}{\frac{nS_2^2}{(n-1)\sigma_2^2}} = \frac{\frac{\hat{S}_1^2}{\sigma_1^2}}{\frac{\hat{S}_2^2}{\sigma_2^2}}$$

has the F distribution with $m - 1, n - 1$ degrees of freedom

C.I. for variance ratios

- Let's repeat our scheme again. We denote by $F_{0.01}$ and $F_{0.99}$ the values of F R.V. for which 1% of the area lies in each tail of the F distribution, then with 98% confidence we have the following

$$F_{0.01} \leq \frac{\frac{\hat{S}_1^2}{\sigma_1^2}}{\frac{\hat{S}_2^2}{\sigma_2^2}} \leq F_{0.99}$$

- So, a 98% C.I. for the variance ratio σ_1^2/σ_2^2 of two populations is given by:

$$\frac{1}{F_{0.99}} \frac{\hat{S}_1^2}{\hat{S}_2^2} \leq \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{1}{F_{0.01}} \frac{\hat{S}_1^2}{\hat{S}_2^2}$$

- The respective critical values are tabulated. The way we read the tables are a bit difficult but we do that during the tutorial, so don't worry!

Examples – C.I.

- **Ex.** Two samples of sizes 16 and 10 are drawn from two normal populations. If their variances are found to be 24 and 18, respectively, find (a) 98% and (b) 90% confidence limits for the ration of the variances

✓ First we can find the modified sample variances:

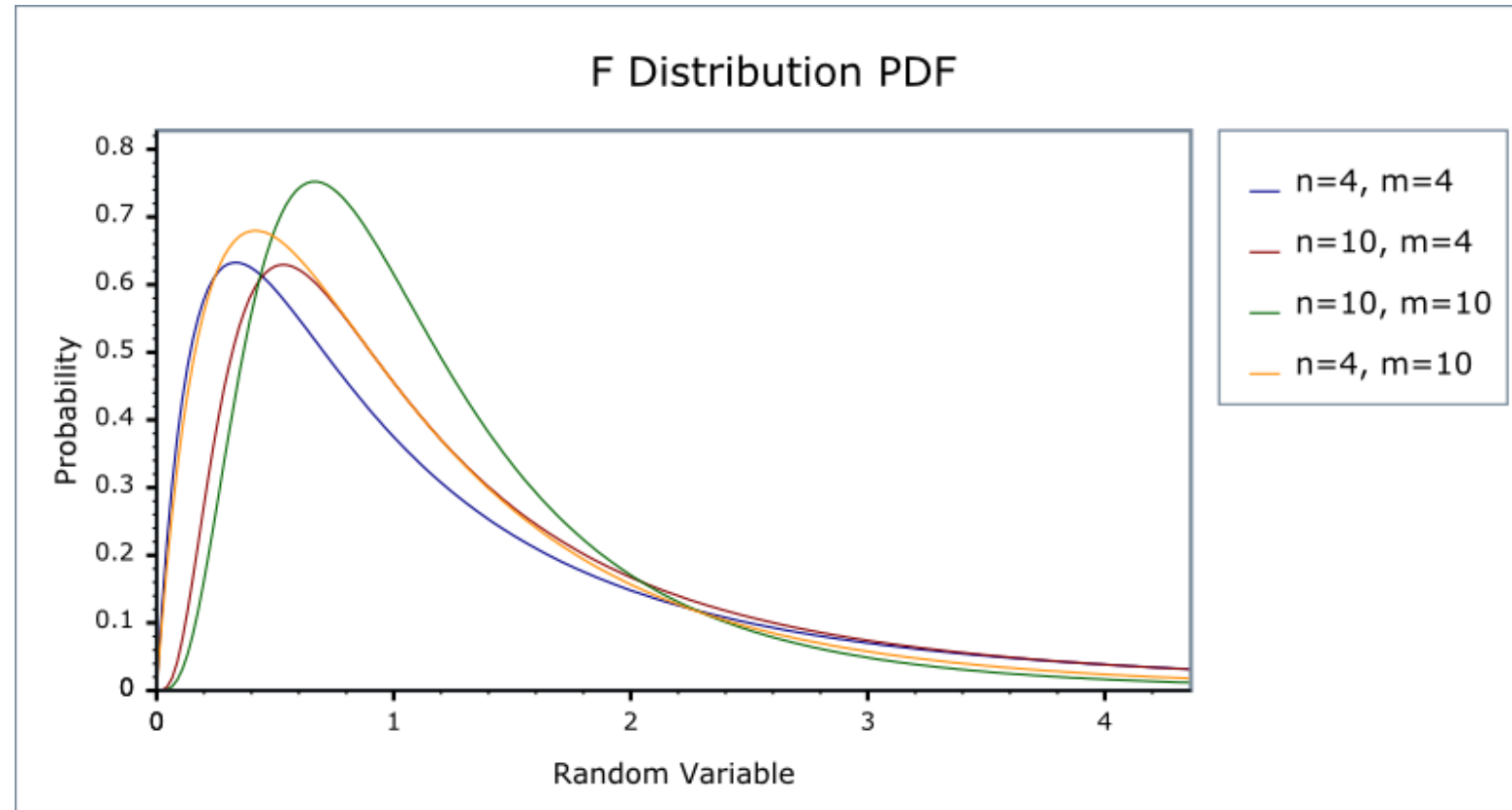
$$\hat{s}_1^2 = \frac{m}{m-1} s_1^2 = \frac{16}{15} \cdot 24 = 25.2$$

$$\hat{s}_2^2 = \frac{n}{n-1} s_2^2 = \frac{10}{9} \cdot 18 = 20$$

- ✓ We need to use F distribution. For the 98% confidence interval we have: $F_{0.99} = 4.96$ for $v_1 = 16 - 1 = 15$ and $v_2 = 10 - 1 = 9$. And $F_{0.01} = 1/3.89$ for the same values of degrees of freedom. So, the c.i.:

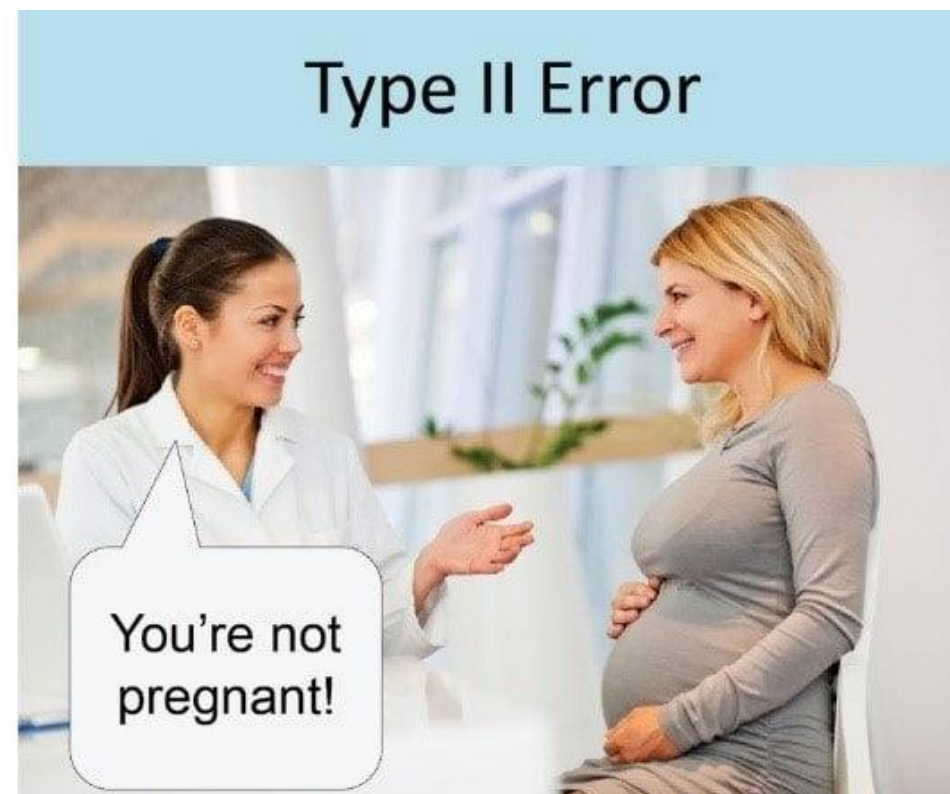
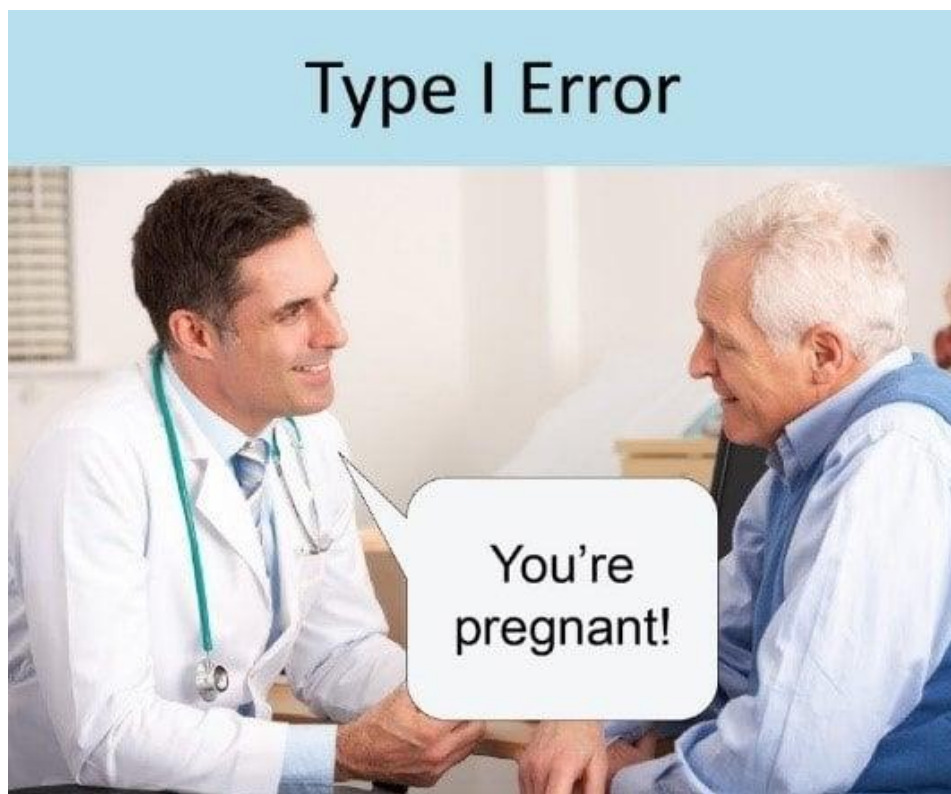
$$C.I._{0.98}^{s^2} = \frac{1}{4.96} \frac{25.2}{20} < \frac{\sigma_1^2}{\sigma_2^2} < 3.89 \frac{25.2}{20} = 0.28 < \frac{\sigma_1^2}{\sigma_2^2} < 4.90$$

Examples – C.I.

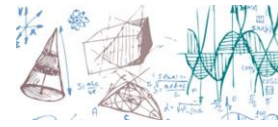


Type I and type II error discussion

H_0 : negative test result



Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality



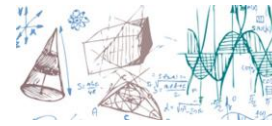
What can happen?

- No matter what we do, in real life we are never going to know for certain that we made the right choice or we in fact made one of the errors
- Remember Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality
- Let's start from a generic (and easy) example, then we try to make a number of general statements and we get back to some more examples...

- Say, we are testing (for normal distribution) the following:

$$H_0: \mu = 14 \quad H_1: \mu > 14$$

- After applying our „procedure” we decided to reject the hypo at significance level $\alpha = 0.05$
- Either of the two happened: **the null is wrong** and we did good or the **null is true** and we made a Type I error



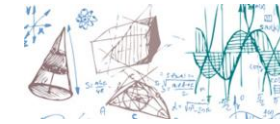
What can happen?

- After applying our „procedure” we decided to accept the hypo at significance level $\alpha = 0.05$
- Either of the two happened: **the null is ok** and we make a good decision or the **null is false** and we made a Type II error
- Is it a bit confusing...? Yes, and we can summarise this using the CONFUSION MATRIX – now you know why we use this name...

Ground truth		
	H_0 is false	H_0 is true
Conclusions using data →	Reject	Type I error
	Do not reject	Type II error

Note, using data we in fact make calculations, the ground truth remains, however, unknown...

A trial example



- This is a classic example: criminal trial

H_0 : *not guilty (no crime)*,

H_1 : *guilty (criminal)*

- In a typical jurisdiction system we have a rule „not guilty until proven otherwise”, so we need to have very strong evidences to convict
 - ✓ Type I: *rule guilty when in reality a person is innocent*
 - ✓ Type II: *not guilty when a person committed a crime*
- Definitely, we want to make sure, that we never convict an innocent person (well, in real life we do...)
- What we can do – we should decrease the possibility of making Type I error as much as possible

A trial example

H_0 : not guilty (no crime),

H_1 : guilty (criminal)

decision	truth	
	guilty	not guilty
not guilty	Type II error	correct
guilty	correct	Type I error



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- Ok, it is time to write down some equations!

Significance and power

Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality

- The probability of **Type I error** is called the **significance level** of the statistical test (we know that already), we write this as

$$P(\text{Type I error} | H_0) = \alpha$$

- Usually we start the test procedure by setting the value of α
- Now, we call the probability of making a Type II error as β
 - ✓ There is a complication – it is much harder to define this error, it depend on the **true** value of the testing parameter, the size of the data sample and on the significance level itself!
- The **POWER** of a statistical test is **the probability of rejecting the null hypothesis when it is false**, we write

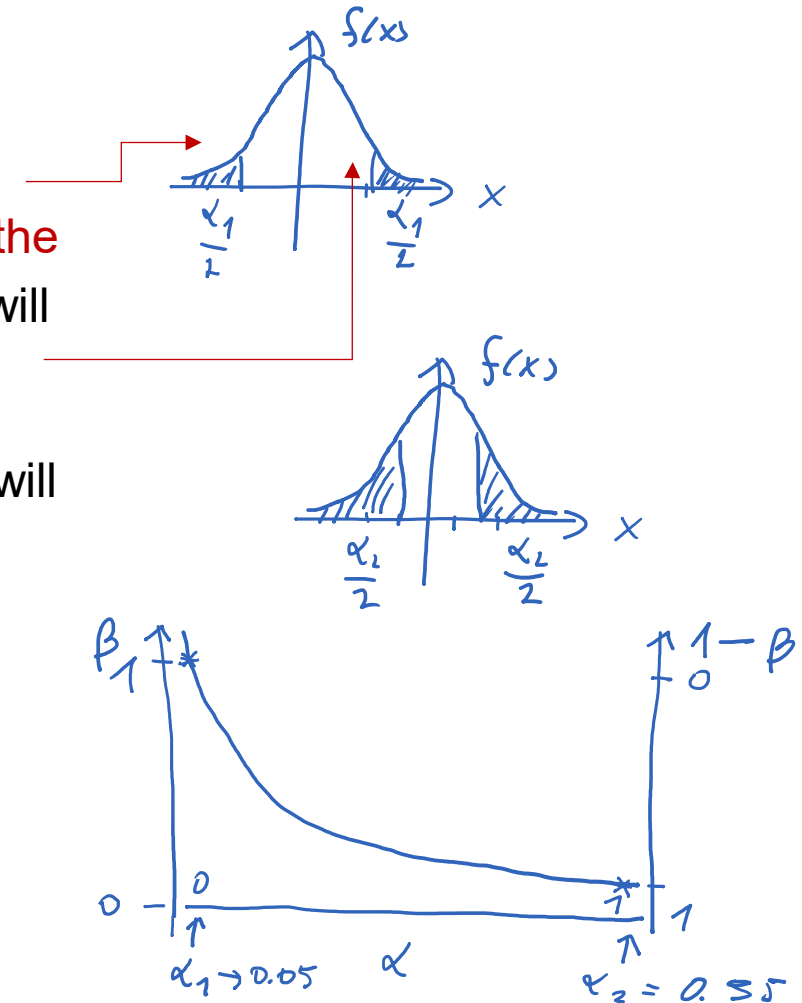
$$\mathcal{R} = 1 - P(\text{Type II error}) = 1 - \beta$$

- The power depends on β – and we still do not know how to calculate it...

Interplay of α and β

Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality

- Ok, we still build up the suspense... Before we show all there is about β let's get some intuitive understanding
- So, why not pick-up some very, very **tiny** value for the significance? The critical region would be very small, thus we would make almost impossible to reject the null hypo. We also say, **that the efficiency of the selection is very high**, but the probability of making the Type II error will also be very high (or we say, **the purity of the selection is low**)
- Conversely, for the large values of α the probability of Type II errors will go down
- Also, from this discussion you should start to understand, that the dependency is not trivial (i.e., it is not just some linear function or such)



Examples!

Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality

- **Ex. 1** Say, we consider a sampling from a normal distribution with known variance $\sigma = 18$, the size of the data sample $n = 36$. Our hypoteses

$$H_0: \mu = 40, \quad H_1: \mu < 40$$

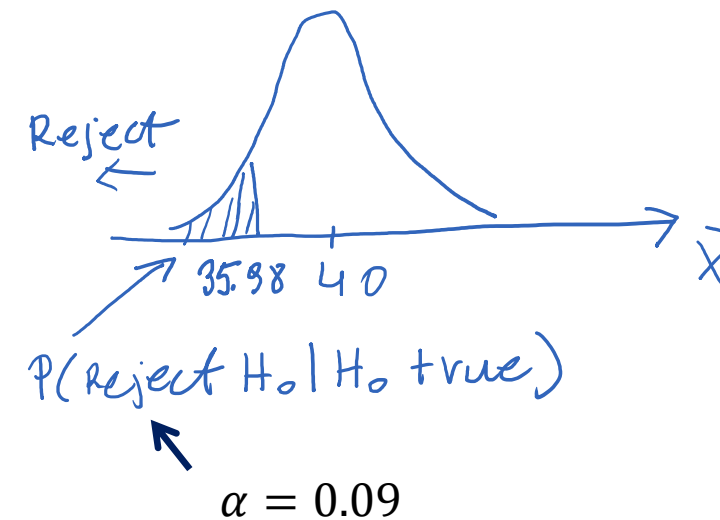
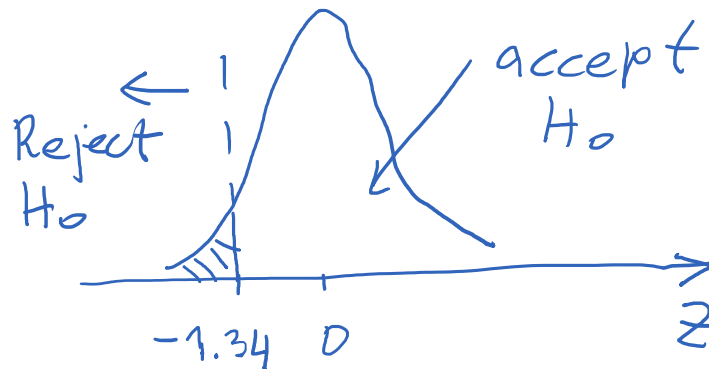
- We set the significance of the test: $\alpha = 0.09$ and our test statistics is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$$

$$\bar{X} = \mu_0 + Z \frac{\sigma}{\sqrt{n}} = 40 + \frac{18}{\sqrt{36}}(-1.34)$$

- The critical region

$$\bar{X} = 35.98$$



Examples!

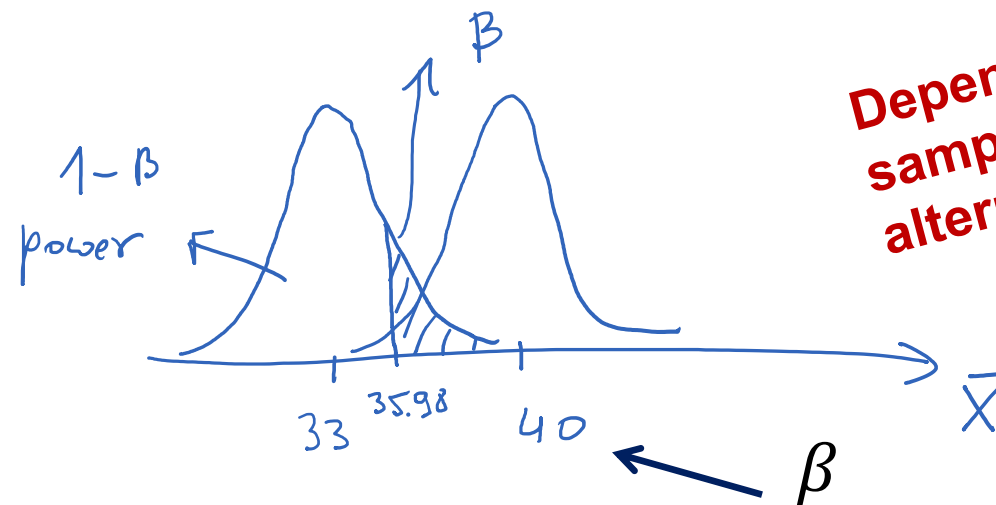
Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality

- Now, what if the true mean is $\mu = 33$? What is the Type II error in this case?

$$P(\text{Accept } H_0 | \mu = 33)$$

- We reject the H_0 hypo if the mean is less than 35.98, so the Type II error will occur with the probability

$$P(\bar{X} > 35.98 | H_1) = P(\bar{X} > 35.98 | \mu = 33)$$

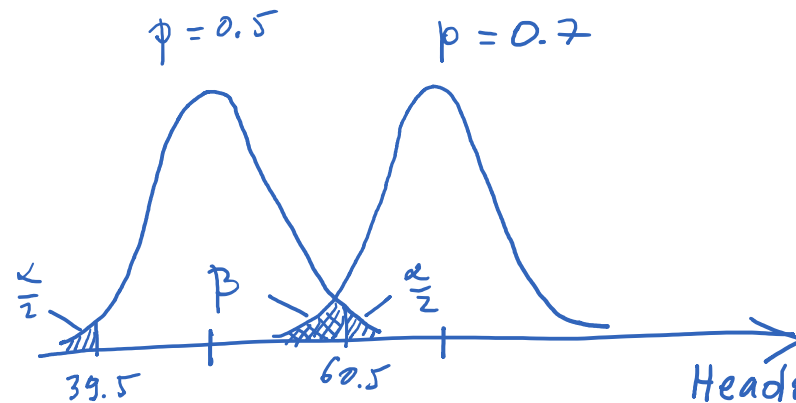


$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{35.98 - 33}{18/6} = \frac{2.98}{3} = 0.99 \rightarrow P(Z > 0.99 | H_1) = 0.16$$

Examples!

Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality

- Note, that when we calculate power and probability of the Type II error we use the alternative hypothesis – we say the calculations are done under the alternative hypothesis
- **Ex. 2** Let's go back to the example with the coin (last lecture) we mentioned the possibility of accepting it if the true probability of heads is $p = 0.7$. This time we calculate the exact value of the probability of accepting the null hypo (i.e., the coin is fair)
 - ✓ Our hypos are: $H_0: p = 0.5, H_1: p = 0.7$
- The numbers we worked out: for the sample of **100 tosses**, we accept the null if we get between **39.5 and 60.5 heads**. This gave us, in turn, the significance $\alpha = 0.0358$.



Examples!

Type I is to **REJECT** the null when it is true and Type II is **NOT TO REJECT** the null when it is false in reality

- The right-hand side distribution represent 100 tosses with the probability of getting head set to 0.7
- Let's work out the probability of the Type II error

$$\mu_1 = np = 100 \cdot 0.7 = 70, \sigma_1 = \sqrt{npq} = 4.58$$

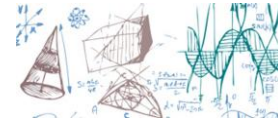
- Now, we need to translate the critical points of the null hypo into Z-score for the alternative hypo

$$z_{+/-}^{(1)} = \frac{h_{+/-} - \mu_1}{\sigma_1} \rightarrow z_+^{(1)} = \frac{60.5 - 70}{4.58} = -2.07, z_-^{(1)} = \dots = -6.66$$

$$\beta = P(-6.66 \leq z_1 \leq -2.07 | H_1) = 0.019$$

- There are slim chances, we accept the null (the coin is fair) when it is loaded with the probability of getting heads 0.7
- There is more to this example... When discussing the *C.I.* we just picked two values for heads and checked the probability – from this we were able to get the significance level – **we say the decision rule was given**

Examples!



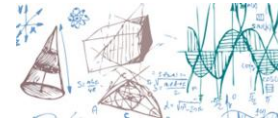
- We can also chose other strategies
 - ✓ **We could decide on the significance level** (the most popular way) and work out the decision rule and the Type II error
 - ✓ **We choose both α and β** and then get a decision rule
- **Ex. 3** Parametric scan of Type II and the test power as a function of heads probability – for the sake of this example we choose: $p = 0.4, 0.6, 0.8, 0.9$ (H_2, H_3, H_4, H_5)
- Can do the calculations as previously: evaluate the μ_i, σ_i first and then $z_{+/-}^{(i)}$ using the critical points of the null hypo as:

$$\mu_1 = 40, \sigma_1 = 4.90, z_{-}^{(2)} = \frac{39.5 - 40}{4.90} = -0.10, z_{+}^{(2)} = 4.18$$

$$\mu_2 = 60, \sigma_2 = 4.90, z_{-}^{(2)} = \frac{39.5 - 60}{4.90} = -4.18, z_{+}^{(2)} = 0.10$$

$$\mu_3 = 80, \sigma_3 = 4.0, z_{-}^{(3)} = \frac{39.5 - 80}{4.90} = -10.12, z_{+}^{(2)} = -4.88$$

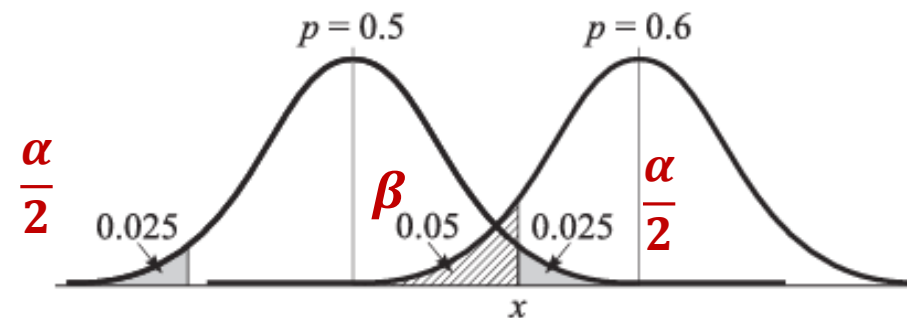
Evaluating sample size



- **Ex. 5** Back again to the coin example, this time we want to work out the sample size given the null and alternative and both error types.

Our null will be $H_0: p = 0.5$ and alternative $H_1: p \neq 0.5$. This time, however, we set that the max difference can be $\Delta p = 0.1$. The $\alpha = 0.05$ at the most and $\beta = 0.05$ at the most. What is $n_{min} = ?$

- The restrictions coming from the limits set on the errors can be represented using graphs, here x is the number of heads observed



- We need to evaluate: $P(x > x_c | H_0) = \frac{\alpha}{2}$, $P(x < x_c | H_1) = \beta$

Evaluating sample size



- We have the following:
$$\frac{x_c - np_0}{\sqrt{np_0q}} = \frac{x_c - 0.5 \cdot n}{0.5 \cdot \sqrt{n}} \rightarrow \frac{\alpha}{2}$$

$$\frac{x_c - np_1}{\sqrt{np_1q}} = \frac{x_c - 0.6 \cdot n}{0.49 \cdot \sqrt{n}} \rightarrow \beta$$

- Note, by making these assumptions we automatically assert that the max Type II error for cases where the p is outside the range (0.4, 0.6) will be equal or smaller than 0.05! Using the normal approximations we have

$$\frac{x_c - 0.5 \cdot n}{0.5 \cdot \sqrt{n}} = 1.96 \rightarrow x_c = 0.5 \cdot n + 0.98 \cdot \sqrt{n}$$

$$\frac{x_c - 0.6 \cdot n}{0.49 \cdot \sqrt{n}} = -1.645 \rightarrow x_c = 0.6 \cdot n - 0.806 \cdot \sqrt{n}$$

$$n = 318.98 \rightarrow n_{min} = 319, x_c = 177$$

- Accept the null hypo if the number of heads are in range: 159.5 ± 17.5